Task - 10: Firewall Configuration & Testing

Internship Task – Elevate Labs

This task focused on understanding firewall concepts and configuring firewall rules to control network traffic. The objective was to learn how firewalls protect systems by allowing or blocking connections based on security rules.

1. Introduction to Firewalls

A firewall is a security mechanism that monitors and controls incoming and outgoing network traffic based on predefined rules. Firewalls act as a barrier between trusted and untrusted networks.

2. Firewall Concepts

Firewalls can filter traffic based on: - IP addresses - Port numbers - Protocols - Direction (inbound and outbound)

They help reduce the attack surface by restricting unnecessary access.

3. Firewall Configuration

Firewall rules were configured using tools such as UFW on Linux or Windows Firewall. Rules were created to allow or deny traffic based on security requirements.

4. Allowing and Blocking Ports

Specific ports were allowed for required services and blocked for unused or risky services. This helps prevent attackers from accessing vulnerable services.

5. Testing Connectivity

After applying firewall rules, network connectivity was tested to ensure legitimate services were accessible while unauthorized access was blocked.

6. Observing Firewall Logs

Firewall logs were reviewed to monitor allowed and blocked traffic. Logs help in identifying suspicious activity and troubleshooting issues.

7. Blocking Malicious IP Addresses

Malicious or suspicious IP addresses were blocked using firewall rules. This prevents repeated attack attempts from known sources.

8. Documentation of Firewall Rules

All firewall rules and configurations were documented clearly. Proper documentation helps in maintaining and reviewing security policies.

9. Impact Analysis

Firewall configuration improves system security by limiting exposure to threats. Incorrect configuration, however, can disrupt services, highlighting the need for careful rule management.

Learning Outcome

Through this task, I gained practical experience in firewall configuration and testing. I learned how firewalls control network traffic, protect systems, and play a critical role in overall network security.